HRIDAYAM KAPILA

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EDUCATION

Aryabhatta College, University of Delhi — BA Economics, Minor in Mathematics *2022–2026*

First Class with Distinction

SKILLS

Econometrics: Panel Data Analysis (FMOLS, Cointegration, Cross-Sectional Dependence), Time Series (VAR, ARDL, Unit Root Tests — ADF/PP/KPSS), OLS with Newey-West HAC Standard Errors, Granger Causality, Hodrick-Prescott Filtering, Full Model Diagnostics **Data:** Panel Dataset Construction from Multiple Sources (World Bank, IMF, UN datasets), Time Series Collection and Cleaning, Data Validation and Quality Checks **Tools:** R, STATA, EViews, Tableau, LaTeX, Git/GitHub, Excel, Typst **Languages:** Hindi (Native), English (Fluent)

RESEARCH EXPERIENCE

Carbon Emissions in Africa: EKC and Regional Integration *Aug 2025 – May 2026*

• Compiled a balanced panel dataset of 9 African countries (2006–2022) with 15+ environmental, economic, and governance indicators sourced from World Bank, IMF, and UN databases • Applied FMOLS estimator to model long-run cointegration between carbon emissions and regional integration, accounting for cross-sectional dependence, non-stationarity, and endogeneity • Found that trade openness and regional value chains increase consumption-based emissions, while environmental cooperation and institutional integration reduce emissions — government effectiveness emerged as a significant emission-reduction driver • Uncovered a U-shaped Environmental Kuznets Curve using group-mean FMOLS in STATA (contrary to the conventional inverted-U hypothesis), with an approximately one-to-one elasticity between energy use and carbon emissions per capita • Scored Outstanding (10/10) in both semesters; presented at UGREE 2026, University of Delhi

PROJECTS

Econometric Analysis of INR and Exchange Rate (R)

• Sourced monthly INR/USD, CPI, and IIP data from RBI DBIE and NSO/MoSPI — 127 observations, Jan 2014 – Jul 2024 • Constructed a “Growth Gap” variable by applying the Hodrick-Prescott filter to IIP month-over-month growth rates to derive the cyclical component of industrial activity • Built 3 OLS model specifications; conducted diagnostic testing — VIF for multicollinearity, Breusch-Pagan for heteroskedasticity, Durbin-Watson for serial correlation • Applied Newey-West HAC standard errors to correct for autocorrelation and confirm robust inference • Established CPI positively associated with INR depreciation (consistent with Purchasing Power Parity) and Growth Gap with INR appreciation — model explained 94.4% of exchange rate variation

VAR Modeling of Macroeconomic Data (R)

• Conducted unit root tests (ADF, PP, KPSS) across 75 monthly observations of exchange rate, inflation, and output gap to determine order of integration • Tested for cointegration using ARDL bounds test — confirmed absence of a long-run equilibrium relationship • Estimated VAR(1) model with lag order selected via AIC, HQ, SC, and FPE information criteria • Validated model through full diagnostic battery: no serial correlation (p=0.50), no ARCH effects (p=0.44), normally distributed residuals (p=0.25) • Performed Granger causality tests to examine directional predictive relationships between variables

University Rankings Analysis (Tableau)

• Designed interactive Tableau dashboard for visualizing and analyzing global university ranking data

POSITIONS OF RESPONSIBILITY

Co-Founder & VP, Placement Unit (Economics Dept.), Aryabhatta College *Sept 2025 – Apr 2026*

Editor-in-Chief, Mind Over Matter (Economics Society), Aryabhatta College *Oct 2024 – Aug 2025*

Treasurer, Arahata (B.A. Programme Society), Aryabhatta College *Oct 2023 – Oct 2024*